

Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

December 6, 2019

Review Topics

- Probability and Mathematical Logic
- Random Variables
- Discrete PMF and CMF
- Moments
- Continuous Joint Distributions

Probability Problems

- Q. When do I use Bayes' Rule and when do I not?

Probability Problems

- Q. When do I use Bayes' Rule and when do I not?
- A. Bayes Rule is for when you have $P(B|A)$ but you need $P(A|B)$.

Probability Problems

- Q. When do I use Bayes' Rule and when do I not?
- A. Bayes Rule is for when you have $P(B|A)$ but you need $P(A|B)$.
- Example: I know what the chances of an infection given I have the disease, I see an infection and want to know if I have the disease.

Probability Problems

- Q. When do I use Bayes' Rule and when do I not?
- A. Bayes Rule is for when you have $P(B|A)$ but you need $P(A|B)$.
- Example: I know what the chances of an infection given I have the disease, I see an infection and want to know if I have the disease.
- Q. When do I use the Law of Total Probability?

Probability Problems

- Q. When do I use Bayes' Rule and when do I not?
- A. Bayes Rule is for when you have $P(B|A)$ but you need $P(A|B)$.
- Example: I know what the chances of an infection given I have the disease, I see an infection and want to know if I have the disease.
- Q. When do I use the Law of Total Probability?
- A. Law of Total Probability is used when you want to aggregate the probability of an event across some partitioned space.

Probability Problems

- Q. When do I use Bayes' Rule and when do I not?
- A. Bayes Rule is for when you have $P(B|A)$ but you need $P(A|B)$.
- Example: I know what the chances of an infection given I have the disease, I see an infection and want to know if I have the disease.
- Q. When do I use the Law of Total Probability?
- A. Law of Total Probability is used when you want to aggregate the probability of an event across some partitioned space.
- Example: I know that Americans are either Republicans, Independents or Democrats, and I have information about each type of person's behavior, what should I expect?

Boole's Inequality

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) \leq P(A) + P(B)$

Probability Rules: Unions and Intersections

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Probability Rules: Unions and Intersections

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- So if A and B are disjoint: $P(A \cup B) = P(A) + P(B)$.

Probability Rules: Unions and Intersections

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- So if A and B are disjoint: $P(A \cup B) = P(A) + P(B)$.
- Example:
- My dog, Maeby, will only bark at skateboards or salt spreaders. We see skateboards when it is warm, salt spreaders when it is cold.

Probability Rules: Unions and Intersections

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- So if A and B are disjoint: $P(A \cup B) = P(A) + P(B)$.
- Example:
- My dog, Maeby, will only bark at skateboards or salt spreaders. We see skateboards when it is warm, salt spreaders when it is cold.
- We see skateboards with probability .1,
We see salt spreaders with probability .2.

Probability Rules: Unions and Intersections

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- So if A and B are disjoint: $P(A \cup B) = P(A) + P(B)$.
- Example:
- My dog, Maeby, will only bark at skateboards or salt spreaders. We see skateboards when it is warm, salt spreaders when it is cold.
- We see skateboards with probability .1,
We see salt spreaders with probability .2.
- The chance of barking is then $.1 + .2 = .3$

Boole's Inequality

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) \leq P(A) + P(B)$

Probability Rules: Independence

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Probability Rules: Independence

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- If A and B are independent, $P(A|B) = \frac{P(A)P(B)}{P(B)} = P(A)$
- If A is a subset of B, then $P(A|B) = \frac{P(A)}{P(B)}$

Probability Rules: Independence

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- If A and B are independent, $P(A|B) = \frac{P(A)P(B)}{P(B)} = P(A)$
- If A is a subset of B, then $P(A|B) = \frac{P(A)}{P(B)}$
- Example:
- Maeby only wears her boots when there is lots of salt.

Probability Rules: Independence

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- If A and B are independent, $P(A|B) = \frac{P(A)P(B)}{P(B)} = P(A)$
- If A is a subset of B, then $P(A|B) = \frac{P(A)}{P(B)}$
- Example:
 - Maeby only wears her boots when there is lots of salt.
 - There is lots of salt only in occasions there is snow.

Probability Rules: Independence

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- If A and B are independent, $P(A|B) = \frac{P(A)P(B)}{P(B)} = P(A)$
- If A is a subset of B, then $P(A|B) = \frac{P(A)}{P(B)}$
- Example:
 - Maeby only wears her boots when there is lots of salt.
 - There is lots of salt only in occasions there is snow.
 - $P(\text{boots}|\text{snow}) = \frac{P(\text{boots})}{P(\text{Snow})}$



Random Variables

- A random variable is a function that assigns numbers to events.
- A distribution tells us how likely those underlying events are.
- A histogram describes the probability of each discrete outcome.
- A density plot describes the shape of continuous probability distributions.
- Very important to distinguish discrete ($\sum$) random variables from continuous ($\int$) random variables.

Binomial Example of PMF

- Suppose that one out of every five witnesses will support the president.
- We had four witnesses yesterday.
- What is the expected number to support the president?

Binomial Example

$$p(x) = \binom{n}{k} p^k (1-p)^{n-k}$$

If $n = 4$ and $p = .2$ What is $E(X)$?

$$E(X) = \sum_{i \in \{0, 1, 2, 3, 4\}} x p(x)$$

Binomial Example

$$p(x) = \binom{n}{k} p^k (1-p)^{n-k}$$

If $n = 4$ and $p = .2$ What is $E(X)$?

$$E(X) = \sum_{i \in \{0, 1, 2, 3, 4\}} x p(x)$$

$$\begin{aligned} & 0 * \binom{4}{0} p^0 (1-p)^{4-0} + 1 * \binom{4}{1} p^1 (1-p)^{4-1} + \\ & 2 * \binom{4}{2} p^2 (1-p)^{4-2} + 3 * \binom{4}{3} p^3 (1-p)^{4-3} + 4 * \binom{4}{4} p^4 (1-p)^{4-4} = \end{aligned}$$

Binomial Example

$$p(x) = \binom{n}{k} p^k (1-p)^{n-k}$$

If $n = 4$ and $p = .2$ What is $E(X)$?

$$E(X) = \sum_{i \in \{0, 1, 2, 3, 4\}} x p(x)$$

$$\begin{aligned} & 0 * \binom{4}{0} p^0 (1-p)^{4-0} + 1 * \binom{4}{1} p^1 (1-p)^{4-1} + \\ & 2 * \binom{4}{2} p^2 (1-p)^{4-2} + 3 * \binom{4}{3} p^3 (1-p)^{4-3} + 4 * \binom{4}{4} p^4 (1-p)^{4-4} = \\ & 4p(1-p)^3 + 12p^2(1-p)^2 + 12p^3(1-p) + 4p^4 = \end{aligned}$$

Binomial Example

$$p(x) = \binom{n}{k} p^k (1-p)^{n-k}$$

If $n = 4$ and $p = .2$ What is $E(X)$?

$$E(X) = \sum_{i \in \{0, 1, 2, 3, 4\}} x p(x)$$

$$\begin{aligned} & 0 * \binom{4}{0} p^0 (1-p)^{4-0} + 1 * \binom{4}{1} p^1 (1-p)^{4-1} + \\ & 2 * \binom{4}{2} p^2 (1-p)^{4-2} + 3 * \binom{4}{3} p^3 (1-p)^{4-3} + 4 * \binom{4}{4} p^4 (1-p)^{4-4} = \\ & 4p(1-p)^3 + 12p^2(1-p)^2 + 12p^3(1-p) + 4p^4 = \\ & 4 * .2(.8)^3 + 12 * .2^2(.8)^2 + 12 * .2^3(.8) + 4 * .2^4 = 0.8 \end{aligned}$$

Moments

- Archimedes noticed that levers apply torque equal to the product of the length of the lever (r) and the amount of effort (F).

Moments

- Archimedes noticed that levers apply torque equal to the product of the length of the lever (r) and the amount of effort (F).
- Now imagine you are spinning, the spread of your arms matters for how fast you will go and how resistant you are to change.

Moments

- Archimedes noticed that levers apply torque equal to the product of the length of the lever (r) and the amount of effort (F).
- Now imagine you are spinning, the spread of your arms matters for how fast you will go and how resistant you are to change.
- The center of gravity of an object (first moment) vs the swing radius (second moment).

Moments

- Archimedes noticed that levers apply torque equal to the product of the length of the lever (r) and the amount of effort (F).
- Now imagine you are spinning, the spread of your arms matters for how fast you will go and how resistant you are to change.
- The center of gravity of an object (first moment) vs the swing radius (second moment).
 - $E(X) = \int xf(x)dx$ is called the first moment.
 - $E(X^2) = \int x^2f(x)dx$ is the second moment.
 - $E(X^3) = \int x^3f(x)dx$ is the third moment (skewness)
 - $E(X^4) = \int x^4f(x)dx$ is the fourth moment (kurtosis)

Cumulative Poisson Distribution

- Suppose we impeach a president four times in 243 years. $4/243$

Cumulative Poisson Distribution

- Suppose we impeach a president four times in 243 years. $4/243$
- What is the chance of impeaching more than two more in our lifetimes (80 years)?

Cumulative Poisson Distribution

- Suppose we impeach a president four times in 243 years. $4/243$
- What is the chance of impeaching more than two more in our lifetimes (80 years)?
- $\lambda = 4/243 * 80 = 1.3$

Cumulative Poisson Distribution

- Suppose we impeach a president four times in 243 years. $4/243$
- What is the chance of impeaching more than two more in our lifetimes (80 years)?
- $\lambda = 4/243 * 80 = 1.3$
- The CDF of the Poisson distribution is $F(X \leq k) = e^{-\lambda} \sum_{i=0}^k \frac{\lambda^i}{i!}$

Cumulative Poisson Distribution

- Suppose we impeach a president four times in 243 years. $4/243$
- What is the chance of impeaching more than two more in our lifetimes (80 years)?
- $\lambda = 4/243 * 80 = 1.3$
- The CDF of the Poisson distribution is $F(X \leq k) = e^{-\lambda} \sum_{i=0}^k \frac{\lambda^i}{i!}$
- The probability of more than 2 in 80 years is $1 - F(X \leq 2)$

Cumulative Poisson Distribution

- Suppose we impeach a president four times in 243 years. $4/243$
- What is the chance of impeaching more than two more in our lifetimes (80 years)?
- $\lambda = 4/243 * 80 = 1.3$
- The CDF of the Poisson distribution is $F(X \leq k) = e^{-\lambda} \sum_{i=0}^k \frac{\lambda^i}{i!}$
- The probability of more than 2 in 80 years is $1 - F(X \leq 2)$

$$\begin{aligned} &= 1 - e^{-\lambda} \sum_{i=0}^k \frac{\lambda^i}{i!} = 1 - e^{-1.3} \left[1 + 1.3 + \frac{1.3^2}{2} \right] \\ &= 1 - .857 = .143 \end{aligned}$$

Conditional Expectations with One Variable

$$f_{X|X \in A}(x) = \frac{f_X(x)}{P(X \in A)}$$

$$E(X|X \in A) = \int x f_{X|X \in A}(x)$$

So if f_X is the Uniform $(-1, 1)$

- What is $f_{X|X>0}(x)$?
- What is $E(X|X > 0)$?

Example of Joint Density of X and Y

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Find marginal distribution of Y.
- Check whether $f(x, y)$ is a valid pdf.
- Find $E(Y)$
- Find conditional distribution of X given $Y=y$.

Marginal distribution of Y

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral of joint pdf with respect to X.

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Marginal distribution of Y

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral of joint pdf with respect to X.

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f(y) = \int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx$$

Marginal distribution of Y

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral of joint pdf with respect to X.

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f(y) = \int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx$$

$$f(y) = \frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} dx$$

Marginal distribution of Y

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral of joint pdf with respect to X.

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f(y) = \int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx$$

$$f(y) = \frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} dx$$

$$f(y) = \frac{e^{-y}}{y} [-ye^{-x/y}|_{\infty} - -ye^{-x/y}|_0]$$

Marginal distribution of Y

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral of joint pdf with respect to X.

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f(y) = \int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx$$

$$f(y) = \frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} dx$$

$$f(y) = \frac{e^{-y}}{y} [-ye^{-x/y}|_{\infty} - -ye^{-x/y}|_0]$$

$$f(y) = \frac{e^{-y}}{y} [0 + y] = e^{-y}$$

Check if $f(x, y)$ is a pdf.

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral w.r.t. X then Y and see if it is equal to 1.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$$

Check if $f(x, y)$ is a pdf.

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral w.r.t. X then Y and see if it is equal to 1.

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_0^{\infty} \left[\int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx \right] dy \end{aligned}$$

Check if $f(x, y)$ is a pdf.

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral w.r.t. X then Y and see if it is equal to 1.

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_0^{\infty} \left[\int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx \right] dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} dx \right] dy \end{aligned}$$

Check if $f(x, y)$ is a pdf.

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral w.r.t. X then Y and see if it is equal to 1.

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_0^{\infty} \left[\int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx \right] dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} dx \right] dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} [-ye^{-x/y}]_{\infty} - -ye^{-x/y}|_0 \right] dy \end{aligned}$$

Check if $f(x, y)$ is a pdf.

$$f(x, y) = e^{-x/y} \cdot \frac{e^{-y}}{y} \quad \text{if } x, y > 0$$

- Procedure: Take integral w.r.t. X then Y and see if it is equal to 1.

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_0^{\infty} \left[\int_0^{\infty} e^{-x/y} \cdot \frac{e^{-y}}{y} dx \right] dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} \int_0^{\infty} e^{-x/y} dx \right] dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} [-ye^{-x/y}]_{\infty} - -ye^{-x/y}|_0 \right] dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} [0 + y] \right] dy = \int_0^{\infty} e^{-y} dy = 1 \end{aligned}$$

Find $E(Y)$

- Procedure: take integral of values of y times density of y .

$$E(Y) = \int_{-\infty}^{\infty} y \cdot f(y) dy$$

Find $E(Y)$

- Procedure: take integral of values of y times density of y .

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy \\ &= \int_0^{\infty} y \cdot e^{-y} dy \end{aligned}$$

Find $E(Y)$

- Procedure: take integral of values of y times density of y .

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy \\ &= \int_0^{\infty} y \cdot e^{-y} dy \end{aligned}$$

Integrate by parts:

$$\int u dv = uv - \int v du, \quad dv = e^{-y}, \quad v = -e^{-y}, \quad u = y, \quad du = 1$$

Find $E(Y)$

- Procedure: take integral of values of y times density of y .

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy \\ &= \int_0^{\infty} y \cdot e^{-y} dy \end{aligned}$$

Integrate by parts:

$$\int u dv = uv - \int v du, \quad dv = e^{-y}, \quad v = -e^{-y}, \quad u = y, \quad du = 1$$

$$= [-ye^{-y}]_0^{\infty} - \int_0^{\infty} -e^{-y} dy$$

Find $E(Y)$

- Procedure: take integral of values of y times density of y .

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy \\ &= \int_0^{\infty} y \cdot e^{-y} dy \end{aligned}$$

Integrate by parts:

$$\int u dv = uv - \int v du, \quad dv = e^{-y}, \quad v = -e^{-y}, \quad u = y, \quad du = 1$$

$$\begin{aligned} &= [-ye^{-y}]_0^{\infty} - \int_0^{\infty} -e^{-y} dy \\ &= 0 - 0 + 1 = 1 \end{aligned}$$

Find $E(Y)$

- Procedure: take integral of values of y times density of y .

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy \\ &= \int_0^{\infty} y \cdot e^{-y} dy \end{aligned}$$

Integrate by parts:

$$\int u dv = uv - \int v du, \quad dv = e^{-y}, \quad v = -e^{-y}, \quad u = y, \quad du = 1$$

$$= [-ye^{-y}]_0^{\infty} - \int_0^{\infty} -e^{-y} dy$$

$$= 0 - 0 + 1 = 1$$

Alternatively, one could recognize that $Y \sim \text{Exp}(1)$

Find conditional distribution of X given $Y=y$.

- Procedure: use definition of conditional $f(x|y) = \frac{f(x, y)}{f(y)}$

$$f(x|y) = \frac{e^{-x/y} \cdot \frac{e^{-y}}{y}}{e^{-y}}$$

Find conditional distribution of X given $Y=y$.

- Procedure: use definition of conditional $f(x|y) = \frac{f(x, y)}{f(y)}$

$$f(x|y) = \frac{e^{-x/y} \cdot \frac{e^{-y}}{y}}{e^{-y}}$$

$$f(x|y) = \frac{1}{y} \cdot e^{-x/y}$$

